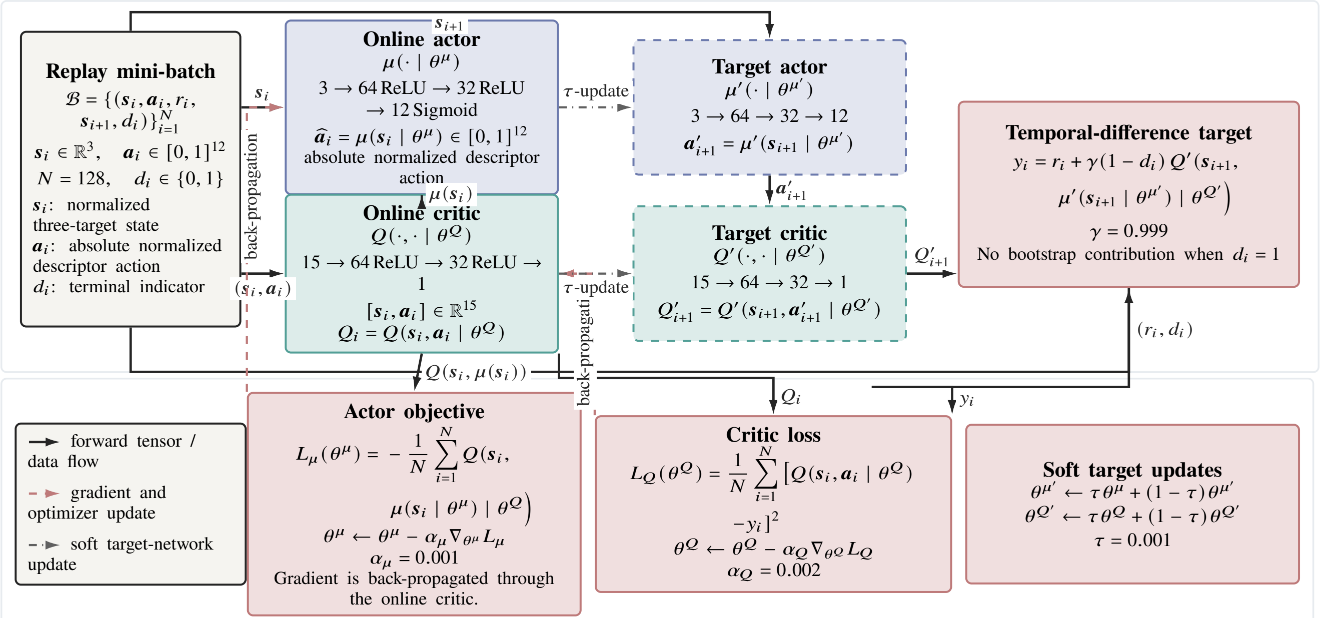
Replay mini-batch
$\mathcal{B} = \{(s_i, a_i, r_i, s_{i+1}, d_i)\}_{i=1}^{N}$
$s_i \in \mathbb{R}^3, \quad a_i \in [0,1]^{12}$
$N = 128, \quad d_i \in \{0,1\}$
$s_i$: normalized three-target state
$a_i$: absolute normalized descriptor action
$d_i$: terminal indicator
Online actor
$\mu(\cdot \mid \theta^{\mu})$
3 → 64 ReLU → 32 ReLU → 12 Sigmoid
$\hat{a}_i = \mu(s_i \mid \theta^{\mu}) \in [0,1]^{12}$
absolute normalized descriptor action
Target actor
$\mu'(\cdot \mid \theta^{\mu'})$
3 → 64 → 32 → 12
$a'_{i+1} = \mu'(s_{i+1} \mid \theta^{\mu'})$
Online critic
$Q(\cdot,\cdot \mid \theta^{Q})$
15 → 64 ReLU → 32 ReLU → 1
$[s_i, a_i] \in \mathbb{R}^{15}$
$Q_i = Q(s_i, a_i \mid \theta^{Q})$
Target critic
$Q'(\cdot,\cdot \mid \theta^{Q'})$
15 → 64 → 32 → 1
$Q'_{i+1} = Q'(s_{i+1}, a'_{i+1} \mid \theta^{Q'})$
Temporal-difference target
$y_i = r_i + \gamma(1-d_i)\, Q'(s_{i+1}, \mu'(s_{i+1} \mid \theta^{\mu'}) \mid \theta^{Q'})$
$\gamma = 0.999$
No bootstrap contribution when $d_i = 1$
$s_{i+1}$
$s_i$
$\tau$-update
$a'_{i+1}$
$\mu(s_i)$
$Q'_{i+1}$
$(s_i, a_i)$
$\tau$-update
back-propagation
back-propagati
$(r_i, d_i)$
$Q(s_i, \mu(s_i))$
$Q_i$
$y_i$
forward tensor / data flow
gradient and optimizer update
soft target-network update
Actor objective
$L_{\mu}(\theta^{\mu}) = -\frac{1}{N}\sum_{i=1}^{N} Q(s_i, \mu(s_i \mid \theta^{\mu}) \mid \theta^{Q})$
$\theta^{\mu} \leftarrow \theta^{\mu} - \alpha_{\mu}\nabla_{\theta^{\mu}} L_{\mu}$
$\alpha_{\mu} = 0.001$
Gradient is back-propagated through the online critic.
Critic loss
$L_Q(\theta^{Q}) = \frac{1}{N}\sum_{i=1}^{N}[Q(s_i, a_i \mid \theta^{Q}) - y_i]^2$
$\theta^{Q} \leftarrow \theta^{Q} - \alpha_{Q}\nabla_{\theta^{Q}} L_Q$
$\alpha_Q = 0.002$
Soft target updates
$\theta^{\mu'} \leftarrow \tau\theta^{\mu} + (1-\tau)\theta^{\mu'}$
$\theta^{Q'} \leftarrow \tau\theta^{Q} + (1-\tau)\theta^{Q'}$
$\tau = 0.001$